

Yan XUE

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Education

- Master in Mathematics, **Università di Padova**, Italy Oct 2022 – July 2025
- **Field of Study:** Mathematics, with a focus on differential geometry
 - **Relevant Coursework:** Differential Geometry, Calculus of Variations, Advance Analysis
- Bachelor of Science, **Southern University of Science and Technology (SUSTech)** Sept 2018 – July 2022
“Double First-Class Construction”, China
- **Field of Study:** Mathematics and Applied Mathematics, with a focus on measure theory

Work Experience

- Data Analyst (Intern)**, Tencent – Shenzhen, China July 2021 – Aug 2021
- Developed Python scripts and web crawlers to automate data collection and cleaning processes.
 - Contributed optimization ideas for user interface and webpage functionality to enhance user experience.
 - Streamlined workflows, reducing manual effort and improving team efficiency.
- Personal Math Tutor**, Self-employed Sept 2018 – Jan 2023
- Provided individualized one-to-one mathematics support, covering topics from secondary-school mathematics to university-level calculus, linear algebra, and operations research.
 - Assisted 35+ university/college and high-school students across diverse academic backgrounds.
- Visiting Scholar**, School of Mathematical Sciences, University of Science and Technology of China (USTC) Apr 2026 – Present
- Attend academic seminars and follow recent developments in relevant research areas.
 - Conduct literature review and theoretical exploration on topics related to supermanifolds, developing and conducting preliminary analysis of several potentially viable theoretical approaches based on existing research.

Selected Technical Projects

- Scientific Visualization & Topological Data Analysis (MATLAB)** Oct 2025
- Developed a MATLAB tool for automated CSV data processing, 3D surface reconstruction, and parameterized visualization of 2D scalar fields.
 - Implemented iso-contour extraction and critical-point detection under specified tolerances, with 2D/3D visualization.
- Numerical Solution & 3D Visualization of the Laplace Equation (Python + ParaView)** Oct 2025
- Implemented a numerical solver for the 2D Laplace equation in Python and automated conversion of results to VTK format for visualization in ParaView.
 - Reconstructed solution surfaces and visualized simulated physical fields in 3D.
- Research of Pendulums with Specific Characteristics (Mathematica)** Sept 2024
- Studied pendulums with oscillating suspension points, periodic length variation, and external forcing.
 - Derived period shift maps, plotted parameter spaces, analyzed fixed points.
 - Designed an algorithm to compute Lyapunov indicators to interpret dynamical behavior.
- Crime Analysis in Washington D.C. (Python)** June 2021
- Conducted data exploration and preprocessing in Python.
 - Analyzed correlations between crime patterns and variables such as time and location using visualization tools.
 - Implemented and evaluated a kNN algorithm for crime pattern analysis, and adapted it to predict housing prices.

Scholarships and Academic Competitions

- Team Leader, China International College Students' Innovation Competition 2025 (CICSIC)
Project name: From Geometry to Physics: Frobenius Structures, Supersymmetry, and Beyond
- Team Member, China International College Students' Innovation Competition 2024 (CICSIC)
Project name: 3D Human Pose and Shape Estimation from RGB Images
- Scholarship for Outstanding Freshmen, SUSTech, 2018

Extracurricular Practical Activities

Volunteer, CKOU South China Kendo Competition	June 2021
Publicity Director and Vice President, Esports Club, SUSTech	Oct 2018 – June 2022

Skills & Interests

Language Skills

- Chinese (native), English (B2-C1), Italian (basic user), Japanese (basic user)

Computer skills

- **Programming & Tools:** Python, C++, Java, SageMath, Mathematica
- **Modeling & Optimization:** MATLAB, AMPL
- **Other Software:** $\LaTeX$, Adobe Photoshop, Adobe Premiere

Interests

- Photography, Kendo, Guzheng, Bass guitar, Rubber stamp sculpture, Rubik's cube, bird watching